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Influence in Correspondence Analysis

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SUMMARY

The influence of individual observations in a correspondence analysis is investigated. Two alternative definitions of 'observation' are possible. For each definition, expressions are presented for influence on the plotting positions of rows and columns of the contingency table being analysed. Formulae for influence on the corresponding eigenvalues (principal inertias) are also given. There are connections for one definition of observation with results for canonical correlation analysis. An example is discussed in detail.

Keywords: Canonical correlation analysis; Contingency tables; Correspondence analysis; Eigenvalues; Eigenvectors; Influence

1. Introduction

The ideas of influence have been explored recently for many of the standard multivariate techniques. For example, Benasseni (1985), Critchley (1985), Pack *et al.* (1988) and Tanaka (1988) all consider aspects of influence in principal component analysis, Campbell (1978) does the same for discriminant analysis and Jolliffe *et al.* (1988) give an empirical study of influence in cluster analysis. Radhakrishnan and Kshirsagar (1981) derived expressions for influence functions for several multivariate procedures, and Tanaka, together with co-workers, has, in a series of technical reports and papers (e.g. Tanaka (1989)) examined several different aspects of 'sensitivity analysis', including detection of influential observations, for a range of multivariate methods.

Despite this activity, little information has yet appeared on influence in correspondence analysis. The main exceptions are Escofier and Le Roux (1976), who investigated the effect of individual columns in the table of data and constructed bounds for the change that occurs when a column is omitted, and Tanaka and Tarumi (1985) who looked at varying the weights of rows in the table of data. Other reports and papers by Tanaka and co-workers consider similar ideas for multiple correspondence analysis.

A first step in investigating influential observations in correspondence analysis is to define what is meant by an 'observation'. Escofier and Le Roux (1976) and Tanaka and Tarumi (1985) considered deletion of a complete row or column from the table of data. However, it is equally valid to think of deleting an observation as corresponding

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to a change in one of the cell frequencies from n_{ij} to $n_{ij}-1$. After introducing notation and terminology in Section 2, influence functions for both types of observation will be discussed in Section 3 of the paper and illustrated by an example in Section 4.

2. Notation and Terminology

Suppose that we have a two-way $I \times J$ contingency table, where the (i,j)th cell frequency is n_{ij}, with marginal totals denoted, in standard notation, by

$$n_{i.} = \sum_{j=1}^{J} n_{ij}, \qquad n_{.j} = \sum_{i=1}^{I} n_{ij}, \qquad n_{..} = \sum_{i=1}^{I} \sum_{j=1}^{J} n_{ij}.$$

In its most usual formulation, the objective of correspondence analysis is to obtain a simultaneous two-dimensional plot of row categories and column categories in the table. Categories which are close to each other in the plot will be those that are closely related to each other according to the cell frequencies of the table.

To construct the plot, an eigenvalue problem must be solved. Define $\mathbf{P}$ to be an $I \times J$ matrix with elements $n_{ij}/n_{..}$, $\mathbf{r}$ and $\mathbf{c}$ respectively to be vectors of marginal totals $\{n_{i.}/n_{..}\}$, $\{n_{.j}/n_{..}\}$, and $\mathbf{D}_r$ and $\mathbf{D}_c$ respectively to be diagonal matrices whose diagonal elements are elements of $\mathbf{r}$ and $\mathbf{c}$. Then the matrices whose eigenvectors we must find are

$$\mathbf{D}_r^{-1}(\mathbf{P}-\mathbf{rc}')\mathbf{D}_c^{-1}(\mathbf{P}-\mathbf{rc}')' \tag{1}$$

and

$$\mathbf{D}_c^{-1}(\mathbf{P}-\mathbf{rc}')'\mathbf{D}_r^{-1}(\mathbf{P}-\mathbf{rc}'). \tag{2}$$

If λ_1 and λ_2 are the two largest eigenvalues of expression (1), then they are also the two largest eigenvalues of expression (2). Denote the corresponding eigenvectors of expressions (1) and (2) respectively by $\mathbf{f}_k$ and $\mathbf{g}_k$, $k=1$, 2, where $\mathbf{f}_k$ and $\mathbf{g}_k$ are normalized so that

$$\mathbf{f}_k'\mathbf{D}_r\mathbf{f}_k = \mathbf{g}_k'\mathbf{D}_c\mathbf{g}_k = \lambda_k. \tag{3}$$

Then the elements of $\mathbf{f}_k$ and $\mathbf{g}_k$ give respectively co-ordinates for the row and column categories with respect to the kth axis of the two-dimensional plot. In practice, we do not find $\mathbf{f}_k$ and $\mathbf{g}_k$ directly as eigenvectors of the non-symmetric matrices (1) and (2). The most useful ways of proceeding are by a singular value decomposition of $\mathbf{D}_r^{-1/2}(\mathbf{P}-\mathbf{rc}')\mathbf{D}_c^{-1/2}$, or by symmetrizing expressions (1) and (2) as outlined in Appendix A.

It will be convenient here to use the symmetric normalizations (3). The distances between row points (column points) then approximate the χ^2-distances between the rows (columns). However, for the distance between a row point and a column point to have a sensible interpretation, *one* of the two normalizations must be changed—see, for example, Greenacre (1984), section 4.6.1.

3. Influence Functions in Correspondence Analysis

As mentioned in Section 1, we can distinguish two types of observation in a contingency table. Perhaps the most obvious definition is simply any one of the observations which are classified in the table; addition or deletion of one extra observation leads to a change in one of the cell frequencies from n_{ij} to $n_{ij}+1$ or $n_{ij}-1$. An alternative type of observation is a complete row or column of the table.

For either type of observation, we can examine how much 'influence' an observation has by repeating the analysis with and without the observation and comparing the two sets of results. However, it would be useful not to have to repeat this simple, but lengthy, procedure for every potential observation in each new data set. Instead, we can derive a theoretical influence function leading to a formula which can be used to estimate the influence of any observation in any data set.

The idea of the theoretical influence curve or theoretical influence function due to Hampel (1974) has been widely used—see, for example, the references in Section 1. The results are usually quoted in terms of *adding* an observation to a distribution, and this convention is followed here. Deletion of an observation can be dealt with in an analogous manner. We shall use the notation $\mathrm{TIC}(\mathbf{x}, \boldsymbol{\theta})$ to denote the theoretical influence function for the vector of parameters $\boldsymbol{\theta}$, evaluated for an observation $\mathbf{x}$. In the present context, we shall be interested in the influence of individual observations on the row and column co-ordinates in the two-dimensional plot, i.e. $\mathrm{TIC}(\mathbf{x}, \mathbf{f}_k)$ and $\mathrm{TIC}(\mathbf{x}, \mathbf{g}_k)$. In addition, $\mathrm{TIC}(\mathbf{x}, \lambda_k)$ will be examined, because λ_k measures how much of the total variation (of cell frequencies about the 'independence' hypothesis) is accounted for by each dimension in the plot.

3.1. *Influence Functions when a Single Observation is Added to (i, j)th Cell*

Perhaps the easiest way to derive theoretical influence functions for adding a single observation in correspondence analysis is to use the fact that correspondence analysis can be viewed as a special case of canonical correlation analysis (see Greenacre (1984), pages 108–116). Expressions for influence functions for the latter technique can then be specialized to give formulae for correspondence analysis.

Consider an $I \times J$ contingency table, with notation as in Section 2, and define, for each of the $n_{..}$ observations in the table, two random variables $\mathbf{y}$ and $\mathbf{x}$ with J and I elements respectively. For an observation in the (i, j)th cell of the table, the ith element of $\mathbf{x}$ and the jth element of $\mathbf{y}$ are 1, with all other elements of $\mathbf{x}$ and $\mathbf{y}$ equal to 0.

It can be shown (an outline proof is given in Appendix A) that the theoretical influence functions which we require have the following form:

$$\mathrm{TIC}(\mathbf{x}, \mathbf{y}, \lambda_k) = -f_{ki}^2 + \frac{2}{\lambda_k^{1/2}} f_{ki} g_{kj} - g_{kj}^2 \tag{4}$$

and

$$\begin{aligned}\mathrm{TIC}(\mathbf{x}, \mathbf{y}, \mathbf{g}_k) = {} & \frac{1}{2\lambda_k}\left(-f_{ki}^2 + \frac{2}{\lambda_k^{1/2}} f_{ki} g_{kj} - g_{kj}^2\right)\mathbf{g}_k + \frac{1}{2}\mathbf{g}_k - \frac{1}{2}\mathbf{D}_c^{-1}\mathbf{y}\, g_{kj} \\ & - \sum_{\substack{t=1\\ t\neq k}}^{J} \frac{\mathbf{g}_t}{\lambda_t}(\lambda_t - \lambda_k)^{-1}\Bigg[-(\lambda_t^{1/2} f_{ti} - g_{tj})(\lambda_k^{1/2} f_{ki} - g_{kj}) \\ & + \left\{1 - \frac{1}{2}(\lambda_k + \lambda_t)\right\} g_{tj} g_{kj}\Bigg].\end{aligned} \tag{5}$$

A similar expression can be found for $\mathrm{TIC}(\mathbf{x}, \mathbf{y}, f_k)$.

3.2. *Influence Functions When a Row is Added to an $I \times J$ Table with $I > J$*

When a row rather than a single observation is added to a contingency table, the approach adopted in Section 3.1 can no longer be used. Instead, formulae have been derived by considering the change in the sample quantities when we add a row consisting of m observations. This change is examined as $n_{..} \to \infty$, or defining $\epsilon = (n_{..}+m)^{-1}$, as $\epsilon \to 0$. By letting $\epsilon \to 0$, we can derive an influence function, which we denote by EIC. Despite the terminology (the E stands for our empirical approach), this is not the same as the empirical influence curve described by Critchley (1985). It also differs from the deleted empirical influence curve and the sample influence curve. However, it seems perhaps the most obvious approach in the present case. The fact that Critchley (1985) noted asymptotic agreement between empirical, deleted empirical and sample curves, coupled with our consideration of $n_{..} \to \infty$, suggests that alternative approaches will lead to similar results. A referee has suggested that $m/n_{..}$, rather than m, might be fixed as $n_{..} \to \infty$. This could certainly be of interest, especially if we are considering the *deletion* of an existing row. However, in the context of adding a row and viewing that row as an observation, it is closer to the usual spirit of influence functions if we give that row decreasing weight as $n_{..} \to \infty$.

If we denote our new row by $\mathbf{r}^*$, with jth element r_j^*, we obtain an expression for the influence of the row on λ_k as (see Appendix A)

$$\mathrm{EIC}(\mathbf{r}^*, \lambda_k) = -\sum_{j=1}^{J} r_j^* g_{kj}^2 + \frac{1}{m\lambda_k}\left(\sum_{j=1}^{J} r_j^* g_{kj}\right)^2 \tag{6}$$

$$= -\sum_{j=1}^{J} r_j^* g_{kj}^2 + m f_{ks}^2 \tag{7}$$

where f_{ks} is the plotting position of the extra (supplementary) row with respect to the original $\mathbf{g}_k$ and λ_k. The corresponding expression for $\mathbf{g}_k$ is

$$\mathrm{EIC}(\mathbf{r}^*, \mathbf{g}_k) = -\sum_{\substack{t=1\\t\neq k}} \frac{\lambda_k^{1/2}}{\lambda_t^{1/2}} \mathbf{g}_t(\lambda_t-\lambda_k)^{-1}\left\{-\frac{1}{2\lambda_t^{1/2}\lambda_k^{1/2}}(\lambda_k+\lambda_t)W_{kt} + m f_{ts} f_{ks}\right\}$$
$$+\frac{1}{2}m\mathbf{g}_k - \frac{1}{2}\mathbf{D}_c^{-1}\mathbf{D}_{r^*}\mathbf{g}_k + \frac{1}{2\lambda_k}\mathbf{g}_k(-W_{kk}+m f_{ks}^2) \tag{8}$$

where

$$W_{kt} = \sum_{j=1}^{J} r_j^* g_{tj} g_{kj}.$$

Although this expression is rather complicated, we see that equation (8), like equation (7), is effectively proportional to m.

The formula for $\mathrm{EIC}(\mathbf{r}^*, \mathbf{f}_k)$ has the complication that the perturbed vector has one more element than the original vector, but it can readily be found from equation (8) by using the formula

$$\mathbf{f}_k = \frac{1}{\lambda_k^{1/2}}\mathbf{D}_r^{-1}\mathbf{P}\mathbf{g}_k \tag{9}$$

which links $\mathbf{f}_k$ and $\mathbf{g}_k$—see Calder (1986), pages 179–180. The term in $O(\epsilon^0)$ for the added row is $(1/m\lambda_k^{1/2})\mathbf{r}^{*\prime}\mathbf{g}_k$ which is the formula given by Greenacre (1984) for representing supplementary points on the display.

The expressions given in this section are relevant to the addition of a row to a table in which $I > J$. If we add a column in this case, or add a row when $I < J$, there is the complication that the number of non-trivial dimensions, $\min(I, J) - 1$, is increased by 1. Experience with examples has shown, however, that if we are only interested in the first few (typically only two) dimensions then the formulae of this section (suitably transposed if we are dealing with columns) give good estimates of actual sample change.

4. Example

A single example will be discussed in detail, looking at influence of single observations and at influence of rows and columns. Further examples are considered in Calder (1986), chapter 6, and the results are quantitatively similar.

The example is taken from Greenacre (1984), p. 259, and the table of data is reproduced here as Table 1.

The example is concerned with the worries of Israeli adults according to their country of origin, and that of their father. The plot of the first two dimensions is given in Fig. 1 and the interpretation of the plot is given by Greenacre (1984). The sample size is $n_{..} = 1554$, so we may expect our comparisons of actual sample and estimated change to be good. We shall consider influence in the first two dimensions in detail, since we usually hope that our analysis results in a useful two-dimensional plot, and we shall briefly outline the influence in later dimensions.

4.1. *Adding a Single Observation to a Cell*

4.1.1. *Influence for eigenvalues*

Table 2 gives the cells where the addition of an extra observation will have the

TABLE 1
Contingency table of principal worries of Israeli adults†

Worry	*No. for the following countries of origin:*					*Total*
	ASAF	*EUAM*	*IFAA*	*IFEA*	*IFI*	
ENR	61	104	8	22	5	200
SAB	70	117	9	24	7	227
MIL	97	218	12	28	14	369
POL	32	118	6	28	7	191
ECO	4	11	1	2	1	19
OTH	81	128	14	52	12	287
MTO	20	42	2	6	0	70
PER	104	48	14	16	9	191
Total	469	786	66	178	55	1554

†ENR, enlisted relative; SAB, sabotage; MIL, military situation; POL, political situation; ECO, economic situation; OTH, other; MTO, more than one worry; PER, personal economics; ASAF, Asia/Africa; EUAM, Europe/America; IFAA, Israel, father Asia/Africa; IFEA, Israel, father Europe/America; IFI, Israel, father Israel.

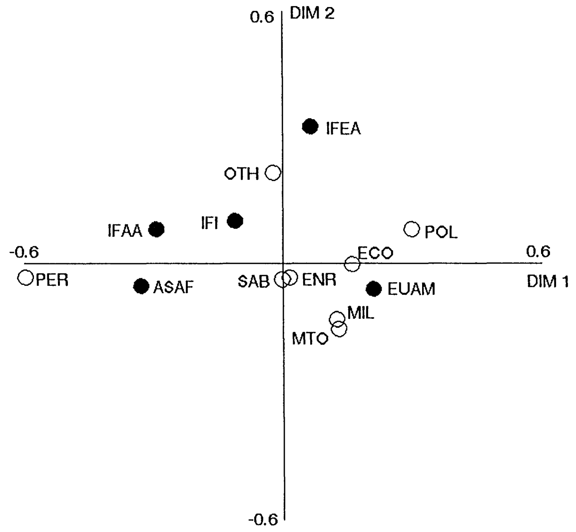

Fig. 1. Plot of the first two dimensions from the correspondence analysis of the data set on worries of Israeli adults: ○, rows; ●, columns

greatest influence on the first two sample eigenvalues $\hat{\lambda}_1$ and $\hat{\lambda}_2$. Table 2 gives the actual sample change, together with the estimated change obtained by substituting sample estimates of λ_k, $\mathbf{f}_k$ and $\mathbf{g}_k$ into equation (4). We shall refer to this estimated change as the empirical influence, in line with Critchley's (1985) terminology. The change is recorded as the percentage change in the eigenvalue.

These changes are small but we are considering a small perturbation (one observation in 1554). The comparisons in Table 2 are good and so we can interpret our asymptotic expression (4), to give us a clear picture of influence in practice. Expression (4) can be re-expressed as

$$\mathrm{TIC}(\mathbf{x}, \lambda_k) = -(g_{kj}-f_{ki})^2 + 2g_{kj}f_{ki}\left(\frac{2}{\lambda_k^{1/2}} - 1\right) \tag{10}$$

where $1/\lambda_k^{1/2} > 1$ as $0 < \lambda_k^{1/2} < 1$. Since we are considering the sample function for adding in points (so the cell entries are incremented by 1) the original parameter is

TABLE 2
Most influential cells (when 1 is added to them) for the first two eigenvalues from the Israeli data set

Cell	*Change in $\hat{\lambda}_1$ (%)*		*Cell*	*Change in $\hat{\lambda}_2$ (%)*	
	Actual	*Estimated*		*Actual*	*Estimated*
8, 2	−1.5	−1.6	6, 4	4.4	4.3
8, 1	1.2	1.2	7, 4	3.5	4.0
8, 3	1.2	0.9	3, 4	3.3	3.5

subtracted from the perturbed. A positive influence thus corresponds to an increase in the parameter.

From equation (10) we obtain a large negative influence, i.e. λ_k decreases, when the co-ordinates g_{kj} and f_{ki} are far apart and have opposing signs. Adding in 1 to the (i, j)th cell increases the association between the corresponding categories. Thus, if g_{kj} and f_{ki} are at the opposite extremes of the dimension they should move inwards towards each other, and so the variance decreases. The eigenvalue increases if g_{kj} and f_{ki} are close and at the extreme of the dimension. This occurs as the row and column are already highly associated with each other, and become more so when 1 is added to the cell entry. Although, as pointed out in Section 2, distances only have a ready interpretation if taken between pairs of rows or of columns, the behaviour just noted shows that relative positions of rows and columns, which we represent by the transition formula (9), also make sense intuitively.

The comparisons between sample and empirical functions were good in this data set, so we find that the above patterns are exhibited for this contingency table. The influence of cell (8, 2) on $\hat{\lambda}_1$ is negative and from Fig. 1 the co-ordinates for 'personal economics' (PER) and 'Europe and America' (EUAM) are far apart at opposing sides of the axis. Conversely, the co-ordinates for PER and 'Asia/Africa' (ASAF) are close together and both large, and the effect of cell (8, 1) on $\hat{\lambda}_1$ is positive. The same pattern occurs for the second axis with, for example, the effect of cell (6, 4) being positive and the co-ordinates for the categories 'other' (OTH) and 'Israel, father Europe/America' (IFEA) being close together and large in the second dimension.

The comparisons of the empirical and sample functions are better for the early than for the later eigenvalues, but these may be of less interest than the first two dimensions.

There is a tendency for cells with low counts to be most influential on the later eigenvalues, especially when their row and column totals are small as well.

4.1.2. *Influence on co-ordinates (eigenvectors)*

To compare the sample and empirical estimated changes we shall take as our statistic the sum of squares of the individual changes in the **g**- and **f**-co-ordinates for each dimension. An alternative, natural, measure of change weights the squared differences in the elements of **g** and **f** according to the matrices $\mathbf{D}_r$ and $\mathbf{D}_c$ respectively, but we concentrate here on the unweighted measure. Table 3 gives the most influential cells on the **g**- and **f**-co-ordinates for the first two dimensions.

The changes are small partly because they are largely attributable to the observation's effect on just one of the coefficients. The comparisons are very close, especially if we look at the individual changes in the co-ordinates. The comparisons deteriorate for the later dimensions although the order of ranked influences changes little. Although the asymptotic expressions for the influence on the **g**- and **f**-co-ordinates, see for example equation (5), are not easy to interpret, we can obtain a good insight into the influence on the co-ordinates from the following numerical results and plots.

The most influential cells, by the sums of squares of individual coefficient changes, on $\mathbf{f}_1$ and $\mathbf{f}_2$ are dominated by combinations with the row 5 and row 7 categories, i.e. $(5, j)$ or $(7, j)$, $j = 1, 2, \ldots, J$. These rows have the smallest row totals. The column totals are much larger than the row totals and we find that the changes in $\mathbf{g}_1$ and $\mathbf{g}_2$ are

TABLE 3
Most influential observations on the **g**- *and* **f**-*co-ordinates*

	$\mathbf{g}_1$			$\mathbf{g}_2$			$\mathbf{f}_1$			$\mathbf{f}_2$	
Observation	*Sample*	*Empirical*	*Observation*	*Sample*	*Empirical*	*Observation*	*Sample*	*Empirical*	*Observation*	*Sample*	*Empirical*
8, 5	0.0018	0.0018	6, 5	0.0011	0.0012	5, 1	0.0056	0.0062	5, 4	0.0170	0.0194
8, 3	0.0009	0.0010	7, 5	0.0008	0.0010	5, 3	0.0043	0.0050	5, 5	0.0021	0.0019
4, 5	0.0006	0.0006	6, 3	0.0005	0.0005	5, 2	0.0012	0.0014	7, 4	0.0017	0.0017

TABLE 4
Most influential cells on the row co-ordinates from the first dimension of the Israeli worries contingency table†

f_{11}	f_{12}	f_{13}	f_{14}	f_{15}	f_{16}	f_{17}	f_{18}
1, 1 (−0.59)	2, 1 (−0.50)	3, 1 (−0.34)	4, 1 (−0.81)	5, 1 (−7.46)	8, 4 (−0.54)	7, 1 (−2.04)	8, 2 (0.69)
1, 3 (−0.53)	2, 3 (−0.45)	3, 3 (−0.33)	4, 3 (−0.64)	5, 3 (−6.56)	6, 1 (−0.47)	7, 3 (−1.86)	8, 4 (0.50)
1, 2 (0.37)	2, 2 (0.33)	8, 4 (0.32)	4, 5 (−0.25)	5, 2 (3.50)	6, 3 (−0.41)	7, 2 (1.04)	8, 1 (−0.29)
8, 5 (0.22)	2, 5 (−0.17)	3, 2 (0.17)	4, 2 (0.24)	5, 5 (−2.73)	6, 2 (0.32)	7, 5 (−0.90)	8, 3 (−0.26)
1, 5 (−0.21)	8, 5 (0.13)	4, 4 (−0.17)	8, 4 (−0.20)	5, 4 (0.59)	4, 4 (0.22)	8, 5 (0.65)	6, 1 (0.12)

†The actual sample change times 100 is given in parentheses.

less than for $\mathbf{f}_1$ and $\mathbf{f}_2$. Once again, however, the most influential cells on $\mathbf{g}_1$ and $\mathbf{g}_2$ are made up of cells involving the columns with the smallest totals.

Table 4 gives the most influential cells on each of the individual row co-ordinates, in the first dimension, i.e. f_{1i}, $i = 1, \ldots, I$. If the ith row (or jth column) has a small total then the most influential cells on f_{1i} (or g_{1j}) tend to be those involving their own row (column) number. The smaller the row (column) total the larger are the influences. This fact could, to some extent, be anticipated from the formula for χ^2-distances, which involves reciprocals of row and column totals. As well as cell numbers in the ith row often being most influential for f_{1i}, we usually find that there is a pattern to the order of the column numbers. From Table 4 we see this order tends to be $(i, 1)$, $(i, 3)$, $(i, 2)$, $(i, 5)$ and $(i, 4)$. This order coincides with the ranked absolute column co-ordinates in the first dimension, with g_{11} being the largest and g_{14} the smallest; see Fig. 1.

It can be shown that if we add m, rather than 1, observations to a cell, then the empirical influence is simply multiplied by m, i.e. it is additive. To see what happens to sample influence consider Table 5, which gives the five most influential observations on f_{1i} when 10 is added to each cell of the 'Israeli' contingency table. These are almost the same as in Table 4, with the values of the influence being approximately 10 times bigger, although with some small perturbations.

4.2. *Deleting a Row from Table*

It was convenient to present the theory in Section 3.2 in terms of adding a row to a table. However, given an actual table, it does not make much sense to introduce a new row to add; we are far more interested in the influence of the rows that we actually have. The expressions of Section 3.2 can easily be modified to deal with deletion, the main effect being a change in sign of the influence. Thus, a positive influence

TABLE 5
Most influential cells on the row co-ordinates from the first dimension of the Israeli worries contingency table, when 10 is added to each cell†

f_{11}	f_{12}	f_{13}	f_{14}	f_{15}	f_{16}	f_{17}	f_{18}
1, 1 (−0.56)	2, 1 (−0.48)	3, 1 (−0.33)	4, 1 (−0.76)	5, 3 (−9.57)	8, 4 (−0.45)	7, 1 (−1.84)	8, 2 (0.65)
1, 3 (−0.48)	2, 3 (−0.41)	8, 4 (−0.29)	4, 3 (−0.36)	5, 5 (−6.74)	6, 1 (−0.44)	7, 3 (−1.51)	5, 3 (0.58)
1, 2 (0.36)	2, 2 (0.31)	3, 3 (0.25)	4, 2 (0.23)	5, 1 (−5.11)	6, 3 (−0.39)	7, 2 (0.93)	8, 3 (−0.44)
8, 5 (0.26)	5, 3 (0.22)	4, 4 (−0.19)	8, 4 (−0.21)	5, 2 (2.43)	6, 2 (0.30)	8, 5 (0.55)	8, 4 (0.35)
5, 5 (0.25)	8, 5 (0.18)	5, 4 (−0.17)	4, 4 (0.17)	5, 4 (1.24)	4, 4 (0.20)	5, 5 (0.54)	5, 5 (0.32)

†The actual sample change times 10 is given in parentheses.

reflects a decrease in the parameter of interest since we subtract the perturbed from the original.

4.2.1. *Influence on eigenvalues*

The EIC with sample quantities substituted and sample influence functions give similar rankings for the eigenvalues from the first two dimensions of the Israeli data set. The top three most influential rows are given in Table 6. There is a pattern of the empirical (estimated change) underestimating the largest sample (actual change) influences. This pattern of empirical influence underestimating sample influence when sample influences are large has been observed in other contexts too—see Pack *et al.* (1988) for principal component analysis and Williams (1987) for regression. As the influences become smaller the empirical and sample values become closer, as occurs in most applications. The empirical and sample apparently differ quite substantially on the second eigenvalue from the Israeli data set, particularly on

TABLE 6
Most influential observations on the eigenvalues from the first two dimensions of the data set on worries of Israeli adults and their sample and empirical influences

Observation	*Influence (%) for $\lambda_1 = 0.060$*		*Observation*	*Influence (%) for $\lambda_2 = 0.015$*	
	Sample	*Empirical*		*Sample*	*Empirical*
PER	69.7	57.5	OTH	62.0	32.0
OTH	−20.9	−17.2	PER	29.6	−8.8
MIL	−17.4	−14.8	MIL	18.6	8.4

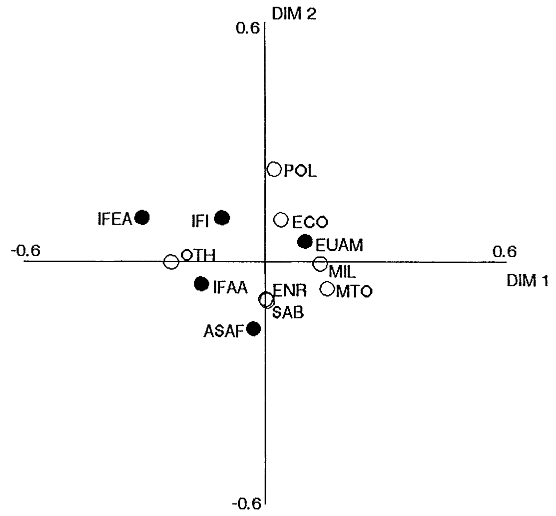

Fig. 2. Plot of the first two dimensions from the correspondence analysis of the data set on worries of Israeli adults when observation PER is deleted: ○, rows; ●, columns

observation PER. This row is highly influential on $\hat{\lambda}_1$, for both functions, and we find when it is omitted that the first two dimensions rotate (and virtually switch); compare Fig. 2 with Fig. 1. This explains why the empirical and sample influences disagree on its influence on $\hat{\lambda}_2$ as the former will not take this rotation, due to close perturbed eigenvalues, into account.

It is the rows with extreme co-ordinates that are the most influential and they have positive influences, which represents a decrease in the variances along the relevant directions. Conversely, points with small co-ordinates have negative influences, which represents an increase in the variance, when they are omitted. Rows with extreme co-ordinates may not be influential if they have a small mass (row sum). This can be seen from the empirical influence function for the ith row which is

$$\frac{1}{n_{..}-m}\mathrm{EIC}(\mathbf{r}^*, \hat{\lambda}_k) = \frac{1}{n_{..}-m}\left(-\sum_{j=1}^{J} r_j^* g_{kj}^2 + m f_{ki}^2\right) \tag{11}$$

where r_j^* is the jth element of the row and m is the row sum.

By comparison the influence measure of Escofier and Le Roux (1976) (see Greenacre (1984), p. 211) is

$$I_{\mathrm{EF}} = -\frac{m}{n_{..}-m}\hat{\lambda}_k + \frac{mn_{..}}{(n_{..}-m)^2} f_{ki}^2. \tag{12}$$

In both equation (11) and equation (12), the second term involving f_{ki}^2 obtains more weight than the first term when m is large, i.e. $n_{..}-m$ is small. The first term in equation (12) is different from that in equation (11) with the former just involving $\hat{\lambda}_k$ and the latter a function of the g-co-ordinates. Expression (12) behaves similarly to the empirical influence but tends to be larger in absolute terms. We also find that, whereas equation (11) is 0 for the trivial dimension, when $f_{ki} = g_{kj} = \lambda_k = 1$ expression (12) is not, but has the value $m^2/(n_{..} - m)^2$.

4.2.2. *Influence on* **g**- *and* **f**-*co-ordinates*

In practice it may be too time consuming to look at all the individual influences on all the **g**- and **f**-co-ordinates. In examples such as the present one, where rotation of axes is a dominant feature, individual changes may also be relatively uninformative. However, some consistent patterns of influence have emerged from examining the individual influences in several examples (see Calder (1986), chapter 6). The most influential rows on the individual $\mathbf{g}_1$-co-ordinates tend to be extreme in the first dimension, but they may be influential on one $\mathbf{g}_1$-co-ordinate and not on another. Whether a row comes out in the top group of influences on a specific g_{ij}-co-ordinate appears to depend jointly on whether it is extreme in the dimension and on the size of its jth residual from the independence model. The signs of influence when extreme rows are taken out are usually the same across all the other row co-ordinates. For example, the signs of influence on the $\mathbf{f}_1$-co-ordinates when an extreme row initially on the right-hand side of the plot is omitted tend to be negative and represent the origin of the axis moving to the left. Another important factor in determining how influential an extreme row is on the other $\mathbf{g}_1$- and $\mathbf{f}_1$-co-ordinates is its mass (row sum). The detailed effect of the mass m is not entirely clear from equation (8), although m appears in several terms, and overall the influence function is roughly proportional to m.

As noted earlier, the rotation caused by the omission of the most influential row PER partially confounds these patterns in the present example. Switching and rotation of dimensions occurs more often in the later dimensions when eigenvalues are small and close, but this example shows that it can also occur in the first few dimensions.

4.2.3. *Scalar measures of influence*

In a practical situation we may not wish to examine the influences on the individual coefficients, because of the numbers involved. We thus require some scalar measure of influence for the **g**- and **f**-co-ordinates in a given dimension, and having decided whether some rows are highly influential we can investigate the individual changes by looking at the perturbed two-dimensional correspondence analysis display. As in Section 4.1.2, one possible scalar measure is the sum of squares of the changes in all the co-ordinates in a given dimension. We need to be careful in the sample case of a change in sign of the perturbed eigenvector. Table 7 gives the most influential rows in the first two dimensions, using the sum of squares of the changes in both the **g**- and the **f**-co-ordinates. The sums of squares are given in parentheses.

An alternative measure of influence, which casts light on possible rotation problems, is the angle between the original and perturbed kth axes. Although the $\mathbf{g}_k$- and $\mathbf{f}_k$-co-ordinates are derived as eigenvectors, they are principal co-ordinates with respect to the principal axes $\mathbf{a}_k$ (in the χ^2-metric D_r^{-1}) and $\mathbf{b}_k$ (in the χ^2-metric D_c^{-1}) respectively. We can thus write

$$\mathbf{g}_k = (\mathbf{D}_c^{-1}\mathbf{P}' - \mathbf{1r}')\mathbf{D}_r^{-1}\mathbf{a}_k \qquad \text{where } \mathbf{a}_k = \frac{1}{\lambda_k^{1/2}}\mathbf{D}_r\mathbf{f}_k,$$

TABLE 7
Most influential observations using the sums of squares of changes in the **f**- *and* **g**-*co-ordinates on the first two dimensions*†

1st dimension		*2nd dimension*	
PER	(0.397)	PER	(0.202)
POL	(0.023)	MIL	(0.044)
MIL	(0.017)	OTH	(0.033)

†The sums of squares of sample changes are given in parentheses.

$$\mathbf{f}_k = (\mathbf{D}_r^{-1}\mathbf{P} - \mathbf{1}\mathbf{c}')\mathbf{D}_c^{-1}\mathbf{b}_k \qquad \text{where } \mathbf{b}_k = \frac{1}{\lambda_k^{1/2}}\mathbf{D}_c\mathbf{g}_k.$$

Details of this are omitted; see Greenacre (1984), pages 88–89. When omitting a row we can find the angle between the original $\mathbf{b}_k$ and the perturbed $\tilde{\mathbf{b}}_k$ (which has an empirical influence function that is similar to that for $\mathbf{g}_k$), but we cannot find the angle for $\mathbf{a}_k$ since we will have vectors of different lengths when we omit a row. However, one angle seems sufficient in indicating the influence of a row. Table 8 gives the most influential observations in the first two dimensions, when using the angular measure of change. The angles are given in parentheses. With either of the scalar measures in Table 7 or Table 8 observation PER stands out as highly influential. It has a larger angle in the second than in the first dimension, indicating that it is not just rotation between these two dimensions that has taken place. We can use the empirical influence function for $\mathbf{b}_k$ to obtain an estimate of the angular change. As with other empirical influence functions, there is a tendency to underestimate the actual angle. Greenacre (1984), pages 213–214, quotes the form of the upper bounds for rotation of the principal axes, derived by Escofier and Le Roux (1976). These were found to be good in the first dimension but poor in the later dimensions.

5. Discussion

Only one example has been discussed here, but it is typical of several which were investigated by Calder (1986). The patterns of influence found reveal in themselves

TABLE 8
Most influential observations using the angular measure of influence for the first two dimensions†

1st dimension		*2nd dimension*	
PER	(48.51°)	PER	(82.31°)
POL	(4.20°)	POL	(10.65°)
MIL	(4.06°)	MIL	(8.37°)

†The sample angular measures are given in parentheses.

that correspondence analysis is a very useful technique in describing the main features of a contingency table. When adding in a single observation to the (i, j)th cell we observed how the ith row and jth column co-ordinates, in the early dimensions, move closer to each other since their association has increased. When deleting an extreme row, columns with large positive and negative residuals with this row move in towards the centre of the display after this strong association has been removed.

The mass of the individual rows and columns is important when determining influences. Rows or columns based on small numbers are understandably the most variable when we add in single observations across their cells. However, if we omit a row with a small mass, even though it may be extreme in the dimension, it has little effect on the plotting positions. This is important as it shows that a correspondence analysis is fairly stable since we can only achieve a large change in the display when an extreme row in the dimension is removed, in the case where that row has a large total. A referee has suggested that this result is fairly obvious because of the nature of the χ^2-distance which implies greater weight for rows or columns with the smallest totals. We agree that some of the results in our example (and others) may be expected from such deductions or from simple intuition. However, our results provide quantitative results for cases where intuition may be a poor guide. The same referee makes the interesting point that a change in one cell will affect *all* χ^2-distances. This contrasts with, for example, principal component analysis, where a change in one observation affects only distances to *that* observation. While on the subject of χ^2-distances, we note that λ_k, $\mathbf{f}_k$ and $\mathbf{g}_k$ may not be the only parameters of interest. We could, for example, investigate the influence of observations on individual χ^2-distances, or on the residual values of these χ^2-distances.

The empirical influence functions derived provide some useful information on what may be influential on the eigenvalues from our correspondence analysis. However, the expressions for the **g**- and **f**-plotting positions were very complex. These expressions would not necessarily be quicker to compute than recalculating the analyses for each row omitted, or cell added to, since they involve summations over all dimensions. However, if we are only interested in the first few dimensions we find that the empirical influences are virtually unchanged if we sum over the first few distinct dimensions only. This occurs as the summation terms involve $(\lambda_j - \lambda_k)^{-1}$ and as λ_j becomes smaller, with λ_k fixed, $(\lambda_j - \lambda_k)^{-1}$ decreases.

The first type of influence analysis involving adding to the (i, j)th cell may only be necessary if the contingency table has some low row totals. This influence analysis would then be examined most simply by a display showing the individual plotting positions when each cell is changed. This would highlight whether any plotting positions were unstable.

When omitting a row it is best to rank the individual influences and to re-form the display for those rows with most influence.

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Appendix A

We here give a brief outline of how the various formulae for influence may be derived. The algebra is somewhat lengthy and will be omitted—further details may be found in Calder (1986).

We have distinguished two cases, where an observation is treated as a simple addition to a single cell of the table and where an observation is considered to be a complete row or column. In the first case, the influence function is derived most conveniently by noting that correspondence analysis can be treated as a special case of canonical correlation analysis.

In canonical correlation analysis, we have two vectors of random variables, $\mathbf{y}$ and $\mathbf{x}$, with p_1 and p_2 elements respectively. The objective is to find linear functions $\mathbf{a}_k'\mathbf{y}$ and $\mathbf{b}_k'\mathbf{x}$ of $\mathbf{y}$ and $\mathbf{x}$ such that $\mathbf{a}_k'\mathbf{y}$ and $\mathbf{b}_k'\mathbf{x}$, called the kth pair of canonical variates, successively have maximum correlation, subject to being uncorrelated with earlier pairs of linear functions. If $\boldsymbol{\Sigma}_{yy}$ and $\boldsymbol{\Sigma}_{xx}$ denote covariance matrices for $\mathbf{y}$ and $\mathbf{x}$ respectively, and $\boldsymbol{\Sigma}_{yx}$ $(=\boldsymbol{\Sigma}_{xy}')$ is a $p_1 \times p_2$ matrix of covariances between elements of $\mathbf{y}$ and $\mathbf{x}$, then the problem is reduced to finding eigenvectors of

$$\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Sigma}_{xy}\boldsymbol{\Sigma}_{yy}^{-1}\boldsymbol{\Sigma}_{yx} \tag{A.1}$$

and

$$\boldsymbol{\Sigma}_{yy}^{-1}\boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Sigma}_{xy}. \tag{A.2}$$

A similarity of structure should be immediately apparent between matrices (A.1) and (A.2), and the matrices relevant to correspondence analysis given earlier in equations (1) and (2).

If $\mathbf{x}$ and $\mathbf{y}$ are defined as in Section 3.1, then canonical correlation analysis reduces to correspondence analysis. Finding the influence functions for correspondence analysis has five steps as follows.

(a) $\boldsymbol{\Sigma}_{yy}^{-1}\boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Sigma}_{xy}$ is converted into a symmetric matrix $\mathbf{S} = \mathbf{C}'\boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Sigma}_{xy}\mathbf{C}$, where $\mathbf{CC}' = \boldsymbol{\Sigma}_{yy}^{-1}$.

(b) We find an expression for the influence of an observation $(\mathbf{x}, \mathbf{y})$ on $\mathbf{S}$.

(c) Expressions are then found for the influence of $(\mathbf{x}, \mathbf{y})$ on the eigenvalues and eigenvectors of $\mathbf{S}$. The methodology for finding influence on eigenvalues and eigenvectors of symmetric matrices is well known—see, for example, Radhakrishnan and Kshirsagar (1981) and Critchley (1985).

(d) Influence functions for the eigenvalues and eigenvectors of $\boldsymbol{\Sigma}_{yy}^{-1}\boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Sigma}_{xy}$ are then derived from those of $\mathbf{S}$.

(e) Finally, we specialize to the case of correspondence analysis. There are two complications, which are readily dealt with, namely different normalization constraints from those most usually adopted in canonical correlation analysis and a trivial first dimension.

For the second case where an observation corresponds to a whole row, we take our original matrix of $\mathbf{x}$ and $\mathbf{y}$ variables and supplement with m extra rows corresponding to the m individuals in the new row of the original table. We can then find $\tilde{\mathbf{D}}$, $\tilde{\mathbf{D}}_c$ and $\tilde{\mathbf{P}}$ for the supplemented $\mathbf{x}$ and $\mathbf{y}$ and express them as an expansion in terms of $\epsilon = (n_{..} + m)^{-1}$ and the original $\mathbf{D}_r$, $\mathbf{D}_c$ and $\mathbf{P}$. An expression for the perturbed value of $\mathbf{C}'\mathbf{P}'\mathbf{D}_r^{-1}\mathbf{PC}$ then follows, and this takes us as far as step (b) in the list above. We can then continue with steps (c)–(e) to obtain our final results.

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